

Panel Data Part 1 and Labeling Data

Samuel Rowe

2025-09-24

Contents

1	Overview	5
2	Panel Data Part 1 - Pooled Data and First Difference	7
2.1	Pooled OLS Estimator	7
2.2	First Difference Estimator	17
3	Labeling Data	41
3.1	Describing datasets	41
3.2	Labeling variables	49
3.3	Labeling values	51
3.4	Labeling utilities	60
3.5	Labeling variables and values in different languages	64
3.6	Using Notes	64
3.7	Formatting the display of variables	66
3.8	Changing the order of variables in a dataset	73
3.9	Practice	76

```
bookdown::serve_book()
```


Chapter 1

Overview

Woolridge Chapter 13 and Mitchell Chapter 5

This week we will start with the implementation of our Pooled OLS estimator and the First Difference Estimator. For data management, we will cover Chapter 5 of Mitchell. This chapter discuss the labeling of data.

1. Pooled OLS Estimator
2. First Difference Estimator
3. Labeling Data

Chapter 2

Panel Data Part 1 - Pooled Data and First Difference

We will cover the implementation of two estimators today: 1. Pooled OLS Estimator 2. First Difference Estimator

2.1 Pooled OLS Estimator

We can account for a_t with our Pooled OLS estimator.

2.1.1 Fertility over time

Lesson: Time binaries can capture secular trends over time or trends that affect all people during the time period.

Sander (1992) uses the National Opinion Research Center's General Social Survey for the veven users from 1972 to 1984. We'll use these data to explain the total number of kids born to a women. Let's get our data.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"  
use "fertil1.dta", clear
```

One question of interest: What happens to ferlity rates over time after controlling for observable factors? We'll control for education, age, race, region at 16 years old, and living environment at 16.

We'll use 1972 as our base year (which we will exclude and be captured in the intercept).

8CHAPTER 2. PANEL DATA PART 1 - POOLED DATA AND FIRST DIFFERENCE

$$kids_{i,t} = \beta_0 + \beta_1 educ_{i,t} + \beta_2 age_{i,t} + \beta_3 age_{i,t}^2 + \beta_4 Black_i + \beta_5 East_{i,t} + \beta_6 NorthCentral_{i,t} \\ + \beta_7 West_{i,t} + \beta_8 Farm_{i,t} + \beta_9 OtherRural_{i,t} + \beta_{10} Town_{i,t} + \beta_{11} SmallCity_{i,t} + a_t$$

First, we'll estimate the regression without Time Trends

```
est clear
```

```
eststo OLS: reg kids educ c.age##c.age i.black i.east i.northcen i.west i.farm i.othru
```

Source	SS	df	MS	Number of obs	=	1,129
Model	314.471892	11	28.5883539	F(11, 1117)	=	11.52
Residual	2771.03741	1,117	2.4807855	Prob > F	=	0.0000
				R-squared	=	0.1019
				Adj R-squared	=	0.0931
Total	3085.5093	1,128	2.73538059	Root MSE	=	1.5751

kids	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
educ	-.1428788	.018351	-7.79	0.000	-.1788851 -.1068725
age	.5624223	.1396257	4.03	0.000	.2884641 .8363804
c.age#c.age	-.0060917	.0015793	-3.86	0.000	-.0091903 -.002993
1.black	.977559	.173188	5.64	0.000	.6377485 1.31737
1.east	.2362931	.1340365	1.76	0.078	-.0266987 .4992849
1.northcen	.3847487	.1222117	3.15	0.002	.1449583 .6245391
1.west	.2447027	.1686052	1.45	0.147	-.0861158 .5755212
1.farm	-.054186	.1486156	-0.36	0.715	-.3457832 .2374112
1.othrural	-.1670751	.1773583	-0.94	0.346	-.5150681 .1809178
1.town	.0842369	.1257038	0.67	0.503	-.1624053 .3308792
1.smcity	.1830768	.1620166	1.13	0.259	-.1348143 .500968
_cons	-8.487543	3.068381	-2.77	0.006	-14.50798 -2.467104

Next, we account for time trends by including Time Fixed Effects

```
eststo Pooled_OLS: reg kids educ c.age##c.age i.black i.east i.northcen i.west i.farm
```

Source	SS	df	MS	Number of obs	=	1,129
Model	399.610888	17	23.5065228	F(17, 1111)	=	9.72
Residual	2685.89841	1,111	2.41755033	Prob > F	=	0.0000
				R-squared	=	0.1295

-----+-----					Adj R-squared	=	0.1162
Total		3085.5093	1,128	2.73538059	Root MSE	=	1.5548
-----+-----							
kids		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+-----							
educ		-.1284268	.0183486	-7.00	0.000	-.1644286	-.092425
age		.5321346	.1383863	3.85	0.000	.2606065	.8036626
c.age#c.age		-.005804	.0015643	-3.71	0.000	-.0088733	-.0027347
1.black		1.075658	.1735356	6.20	0.000	.7351631	1.416152
1.east		.217324	.1327878	1.64	0.102	-.0432192	.4778672
1.northcen		.363114	.1208969	3.00	0.003	.125902	.6003261
1.west		.1976032	.1669134	1.18	0.237	-.1298978	.5251041
1.farm		-.0525575	.14719	-0.36	0.721	-.3413592	.2362443
1.othrural		-.1628537	.175442	-0.93	0.353	-.5070887	.1813814
1.town		.0843532	.124531	0.68	0.498	-.1599893	.3286957
1.smcity		.2118791	.160296	1.32	0.187	-.1026379	.5263961
1.y74		.2681825	.172716	1.55	0.121	-.0707039	.6070689
1.y76		-.0973795	.1790456	-0.54	0.587	-.448685	.2539261
1.y78		-.0686665	.1816837	-0.38	0.706	-.4251483	.2878154
1.y80		-.0713053	.1827707	-0.39	0.697	-.42992	.2873093
1.y82		-.5224842	.1724361	-3.03	0.003	-.8608214	-.184147
1.y84		-.5451661	.1745162	-3.12	0.002	-.8875846	-.2027477
_cons		-7.742457	3.051767	-2.54	0.011	-13.73033	-1.754579
-----+-----							

Exclusion Restriction Test

```
test 1.y74 1.y76 1.y78 1.y80 1.y82 1.y84
```

```
( 1) 1.y74 = 0
( 2) 1.y76 = 0
( 3) 1.y78 = 0
( 4) 1.y80 = 0
( 5) 1.y82 = 0
( 6) 1.y84 = 0
```

```
F( 6, 1111) = 5.87
Prob > F = 0.0000
```

We reject the null hypothesis that our combined time binaries are 0 with an F-stat (1,111 degrees of freedom) equal to 5.87.

Display

```
esttab OLS Pooled_OLS, mtitles se scalars(F r2) drop(0.black 0.east 0.northcen 0.west 0
```

	(1) OLS	(2) Pooled_OLS
educ	-0.143*** (0.0184)	-0.128*** (0.0183)
age	0.562*** (0.140)	0.532*** (0.138)
c.age#c.age	-0.00609*** (0.00158)	-0.00580*** (0.00156)
1.black	0.978*** (0.173)	1.076*** (0.174)
1.east	0.236 (0.134)	0.217 (0.133)
1.northcen	0.385** (0.122)	0.363** (0.121)
1.west	0.245 (0.169)	0.198 (0.167)
1.farm	-0.0542 (0.149)	-0.0526 (0.147)
1.othrural	-0.167 (0.177)	-0.163 (0.175)
1.town	0.0842 (0.126)	0.0844 (0.125)
1.smcity	0.183 (0.162)	0.212 (0.160)
1.y74		0.268 (0.173)
1.y76		-0.0974 (0.179)
1.y78		-0.0687

		(0.182)
1.y80		-0.0713 (0.183)
1.y82		-0.522** (0.172)
1.y84		-0.545** (0.175)
_cons	-8.488** (3.068)	-7.742* (3.052)

N	1129	1129
F	11.52	9.723
r2	0.102	0.130

Standard errors in parentheses

* p<0.05, ** p<0.01, *** p<0.001

We can see most our time coefficient are not statistically significant but we do reject the null hypothesis that they are all equal to 0. If we compare 1982 and 1984 to 1972, women estimated to have about 0.52 fewer and 0.55 fewer children, respectively. This means that in 1982 100 women are expected to have 52 fewer children than 100 women in 1972.

Note: Since we are controlling for education, this decline in fertility cannot be attributed to increases in education. The difference is a secular trend or a general trend that transcends across all women.

Note: We have age and age-squared, which means that the number of children increases with age, but at a diminishing rate. At one point age will become 0.

```
estat hettest
```

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity

Ho: Constant variance

Variables: fitted values of kids

chi2(1) = 26.28

Prob > chi2 = 0.0000

Note that we reject the null hypothesis of homoskedasticity with a Breusch-Pagan Test, so we should use our heteroskedastic-robust standard errors

```

est clear
eststo Without_Robust: quietly reg kids educ c.age##c.age i.black i.east i.northcen i.west
eststo With_Robust: quietly reg kids educ c.age##c.age i.black i.east i.northcen i.west
esttab Without_Robust With_Robust, mtitles se scalars(F r2) drop(0.black 0.east 0.northcen 0.west)

```

	(1) Without_Robust	(2) With_Robust
educ	-0.128*** (0.0183)	-0.128*** (0.0211)
age	0.532*** (0.138)	0.532*** (0.139)
c.age#c.age	-0.00580*** (0.00156)	-0.00580*** (0.00158)
1.black	1.076*** (0.174)	1.076*** (0.201)
1.east	0.217 (0.133)	0.217 (0.127)
1.northcen	0.363** (0.121)	0.363** (0.117)
1.west	0.198 (0.167)	0.198 (0.163)
1.farm	-0.0526 (0.147)	-0.0526 (0.146)
1.othrural	-0.163 (0.175)	-0.163 (0.181)
1.town	0.0844 (0.125)	0.0844 (0.128)
1.smcity	0.212 (0.160)	0.212 (0.154)
1.y74	0.268 (0.173)	0.268 (0.188)
1.y76	-0.0974	-0.0974

	(0.179)	(0.200)
1.y78	-0.0687 (0.182)	-0.0687 (0.198)
1.y80	-0.0713 (0.183)	-0.0713 (0.194)
1.y82	-0.522** (0.172)	-0.522** (0.188)
1.y84	-0.545** (0.175)	-0.545** (0.186)
_cons	-7.742* (3.052)	-7.742* (3.071)

N	1129	1129
F	9.723	10.19
r2	0.130	0.130

Standard errors in parentheses

* p<0.05, ** p<0.01, *** p<0.001

Question: do you think the number of kids is distributed normally?

2.1.2 Changes in the Returns to Education and the Sex Wage Gap

Lesson: Interactions matter. We can manually create interactions, but that can be cumbersome. Using the # operator, we can generate interactions within our regression command.

Stata interactions: *i.var1##i.var2*, *i.var1##c.var3*, *c.var3##c.var4*

We can estimate to see how the wage gap and returns to schooling have compared for women. We pool data from 1978 and 1985 from the Current Population Survey.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use "cps78_85.dta", clear
```

What we will do to estimate the wage gap and the returns to education between 1978 and 1985. We can do this by interacting our binary variable y85 with our continuous variable of education. Our coefficient on edu will be the returns to

education in 1978 and the return to education in 1985 will be captured in the coefficient for `edu#1.y85` plus the coefficient for `edu` in 1978. The coefficient for `edu#1.y85` will be relative to the coefficient on `edu`, so we can add the two coefficients to find the returns to education in 1985.

We also interact our binary variable `y85` with another binary variable `female` to see the wage gap in 1978 and 1985. `1.female` will be the wage gap in 1978 and `1.female#1.y85` plus the `1.female` will be our wage gap for females in 1985.

Wages will have changed between 1978 and 1985 due to inflation, since wages in the CPS are in nominal dollars. We can deflate our wages with a price deflator, or we can use natural log of wages with time binaries to capture inflation that is associated with the year binaries.

Natural Log and accounting for inflation with time variables. Let's say our deflator for 1985 is 1.65, so we need to divide wages by the deflator to get real wages in 1978 dollars. But if we take the natural log transformed wages using real or nominal wages does not matter since the inflation factor will be absorbed in the time binary `y1985`:

$$\ln(wage_i/P85) = \ln(wages_i) - \ln(p85)$$

While wages may differ across individuals in 1985, all individuals in 1985 are affected by the secular trend of inflation relevant to 1985. (Note: if the price trends varied by region, then we would need to account for this since prices would not be a secular trend affecting all people).

We need both $\ln(\text{wages})$ and time binaries, otherwise our estimates would be biased.

Note: that this remains true whether the nominal value is in the dependent or independent variables as long as it is natural log transformed and has time binaries.

$$\ln(wage_{i,t}) = \beta_0 + \beta_1 edu_{i,t} + \beta_2 exp_{i,t} + \beta_3 exp_{i,t}^2 + \beta_4 union_{i,t} + \beta_5 female_i + \beta_6 edu_{i,t} * Y85_t + \beta_7 female_i * Y85_t$$

```
reg lwage i.y85##c.edu c.exper##c.exper i.union i.female##i.y85
```

Source	SS	df	MS	Number of obs	=	1,084
-----+-----				F(8, 1075)	=	99.80
Model	135.992074	8	16.9990092	Prob > F	=	0.0000
Residual	183.099094	1,075	.170324738	R-squared	=	0.4262
-----+-----				Adj R-squared	=	0.4219
Total	319.091167	1,083	.29463635	Root MSE	=	.4127

lwage	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
1.y85	.1178062	.1237817	0.95	0.341	-.125075	.3606874
educ	.0747209	.0066764	11.19	0.000	.0616206	.0878212
y85#c.educ						
1	.0184605	.0093542	1.97	0.049	.000106	.036815
exper	.0295843	.0035673	8.29	0.000	.0225846	.036584
c.exper#c.exper	-.0003994	.0000775	-5.15	0.000	-.0005516	-.0002473
1.union	.2021319	.0302945	6.67	0.000	.1426888	.2615749
1.female	-.3167086	.0366215	-8.65	0.000	-.3885663	-.244851
female#y85						
1 1	.085052	.051309	1.66	0.098	-.0156251	.185729
_cons	.4589329	.0934485	4.91	0.000	.2755707	.642295

Our results:

Wage gap in 1978 is $(e^{(-.317)} - 1) * 100 = -27.2\%$

Wage gap in 1985 is $(e^{(-.317+.085)} - 1) * 100 = -20.7\%$

```
display (exp(-.317)-1)*100
display (exp(-.317+0.085)-1)*100
```

-27.166924

-20.705388

Difference in returns to education

```
display (exp(0.0747)-1)*100
display (exp(0.0747+0.0185)-1)*100
```

7.7560834

9.768125

Declines in wage gap since 1978

```
display (exp(0.085)-1)*100
```

```
8.8717067
```

Increases in returns to education since 1978

```
display (exp(0.0185)-1)*100
```

```
1.8672185
```

Chow Test/F-Test for structural changes across time (nested-model). We will test whether or not to include time binaries.

```
test 1.y85 0.y85 1.y85#edu 1.y85#1.female
```

```
( 1) 1.y85 = 0
( 2) 0b.y85 = 0
( 3) 1.y85#c.educ = 0
( 4) 1.female#1.y85 = 0
      Constraint 2 dropped

      F( 3, 1075) = 80.26
      Prob > F = 0.0000
```

Test slopes

```
test 1.y85#edu 1.y85#1.female
```

```
( 1) 1.y85#c.educ = 0
( 2) 1.female#1.y85 = 0

      F( 2, 1075) = 3.43
      Prob > F = 0.0326
```

Test intercepts

```
test 1.y85 0.y85
```

```
( 1) 1.y85 = 0
( 2) 0b.y85 = 0
      Constraint 2 dropped

      F( 1, 1075) = 0.91
      Prob > F = 0.3415
```


Should we keep our time binaries? Our F-tests show that we should and we'll need them for to account for inflation.

2.2 First Difference Estimator

We can account for a_t and a_i with our First Difference estimator.

2.2.1 Sleep vs Working

Lesson: The way you set up panel data matters. We need to use the long-format.

I think this is a good example of a poorly set up panel data. We'll use the file called `slp75_81.dta` to look at a two year panel data from Biddle and Hamermesh (1990) to estimate the tradeoff between sleeping and working. We have data for 239 people between 1975 and 1981 for the same person in both periods.

We'll get to reshape in Mitchell soon, but here is a preview. We will reshape the data so that each cross-sectional unit of observation has 2 rows: one for 1975 and one for 1985.

Let's take a look at a wide-form dataset that needs to be reshaped

	age75	educ75	educ81	gdh1th75	gdh1th81	male	marr75	marr81	slpnap
1	46	16	16	1	1	1	0	1	
2	39	16	16	1	1	1	1	0	
3	55	15	15	1	0	1	1	1	
4	39	16	16	1	1	1	1	1	
5	54	17	17	1	1	1	1	1	
6	32	16	16	1	1	1	1	1	
7	30	16	16	1	1	1	1	1	
8	27	16	16	0	1	1	0	0	
9	33	17	19	1	1	0	1	1	
10	51	14	14	1	1	0	1	1	
11	23	16	16	1	1	1	1	1	
12	46	17	19	1	1	0	1	1	
13	38	12	12	1	1	1	1	1	
14	30	10	10	1	0	1	0	0	
15	39	12	12	1	0	1	1	1	
16	53	12	12	1	0	1	1	1	
17	59	14	14	1	0	1	1	1	
18	54	14	14	1	1	0	1	1	
19	35	12	12	1	1	0	1	1	
20	46	17	18	1	1	1	1	1	
21	36	12	12	1	1	1	1	1	
22	25	17	18	1	1	1	0	1	
23	29	12	12	1	1	0	0	0	
24	37	12	12	1	0	0	1	1	
25	31	17	17	1	1	1	1	1	
26	43	12	12	1	0	0	1	1	
27	35	12	14	1	1	0	1	1	
28	37	12	14	1	1	1	1	1	
29	28	16	16	1	1	0	0	0	
30	35	12	12	1	0	0	0	0	
31	56	11	12	1	0	1	1	1	

Figure 2.1: Data in wide form

Reshape

18CHAPTER 2. PANEL DATA PART 1 - POOLED DATA AND FIRST DIFFERENCE

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use "slp75_81.dta", clear
gen id=_n
gen age81=age75+6
reshape long age educ gdhlth marr slpnap totwrk yngkid, i(id) j(year)
replace year=year+1900
```

/Users/Sam/Desktop/Econ 645/Data/Wooldridge

(note: j = 75 81)

Data	wide	->	long
Number of obs.	239	->	478
Number of variables	22	->	16
j variable (2 values)		->	year
xij variables:			
	age75 age81	->	age
	educ75 educ81	->	educ
	gdhlth75 gdhlth81	->	gdhlth
	marr75 marr81	->	marr
	slpnap75 slpnap81	->	slpnap
	totwrk75 totwrk81	->	totwrk
	yngkid75 yngkid81	->	yngkid

variable year was byte now int
(478 real changes made)

We'll generate our time binaries

```
gen d75 = .
replace d75=0 if year==1981
replace d75=1 if year==1975

gen d81=.
replace d81=0 if year==1975
replace d81=1 if year==1981
```

(478 missing values generated)

(239 real changes made)

(239 real changes made)

(478 missing values generated)

(239 real changes made)

(239 real changes made)

And we'll want to check that we don't have any overlapping periods Check

```
tab year d75
tab year d81
tab d81 d75
```

		d75		
year		0	1	Total
1975		0	239	239
1981		239	0	239
Total		239	239	478

		d81		
year		0	1	Total
1975		239	0	239
1981		0	239	239
Total		239	239	478

		d75		
d81		0	1	Total
0		0	239	239
1		239	0	239
Total		239	239	478

Now we can work with the panel. Let *xtset* know that there is a 6 year gap in the data. If not, it will try to take a 1 year change and 1976 does not exist.

```
xtset id year, delta(6)
```

```

panel variable:  id (strongly balanced)
time variable:   year, 1975 to 1981
delta:           6 units

```

$$slpnap_{i,t} = \beta_0 + \beta_1 totwrk_{i,t} + \beta_2 educ_{i,t} + \beta_3 marr_{i,t} + \beta_4 yngkid_{i,t} + \beta_5 gdhlth_{i,t} + a_i + a_t$$

1. slpnap - total minutes of sleep per week
2. totwrk - total hours of work per year
3. educ - total years of education
4. marr - married dummy variable
5. yngkid - presence of a young child dummy variable
6. gdhlth - “good health” dummy variable

After we set up our panel, we can control for the unobserved cross-sectional unit effect a_i , or unobserved individual effect that do not vary over time. This is all characteristics of the cross-sectional unit of observation that is constant over time or time-invariant.

We are interested in the tradeoff between work and sleep. Our dependent variable of interest is total minutes sleeping per week. Our explanatory variable of interest is the total working hours during the year.

```

histogram totwrk
graph export "/Users/Sam/Desktop/Econ 645/Stata/totwrk.png", replace

```

```

histogram slpnap
graph export "/Users/Sam/Desktop/Econ 645/Stata/slpnap.png", replace

```

When we run OLS, we do not control for unobserved individual time-invariant effects.

Pooled OLS

```

est clear
eststo OLS: reg slpnap i.d81 totwrk educ marr yngkid gdhlth

```

Source		SS	df	MS	Number of obs	=	478
-----+-----					F(6, 471)	=	8.50
Model		12812772.5	6	2135462.09	Prob > F	=	0.0000

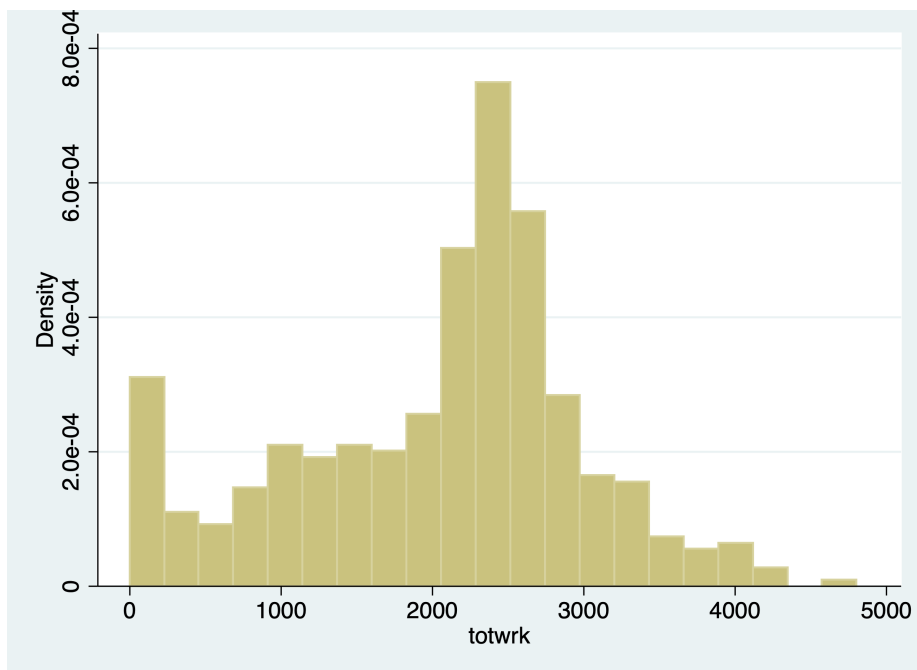


Figure 2.2: Histogram of Annual Hours Worked

Residual		118370944	471	251318.353	R-squared	=	0.0977

Total		131183717	477	275018.274	Adj R-squared	=	0.0862
					Root MSE	=	501.32

slpnap	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
1.d81	-77.02965	47.96519	-1.61	0.109	-171.2819	17.22259
totwrk	-.1500954	.0244002	-6.15	0.000	-.1980421	-.1021487
educ	-20.25655	8.000179	-2.53	0.012	-35.97701	-4.536091
marr	-42.19389	53.07869	-0.79	0.427	-146.4942	62.10645
yngkid	74.2579	77.32582	0.96	0.337	-77.68838	226.2042
gdhlth	29.03733	59.57889	0.49	0.626	-88.03599	146.1107
_cons	3958.756	121.7663	32.51	0.000	3719.483	4198.028

FD Model

```
eststo FD: reg d.slpnap d.totwrk d.educ d.marr d.yngkid d.gdhlth
```

Source		SS	df	MS	Number of obs	=	239
--------	--	----	----	----	---------------	---	-----

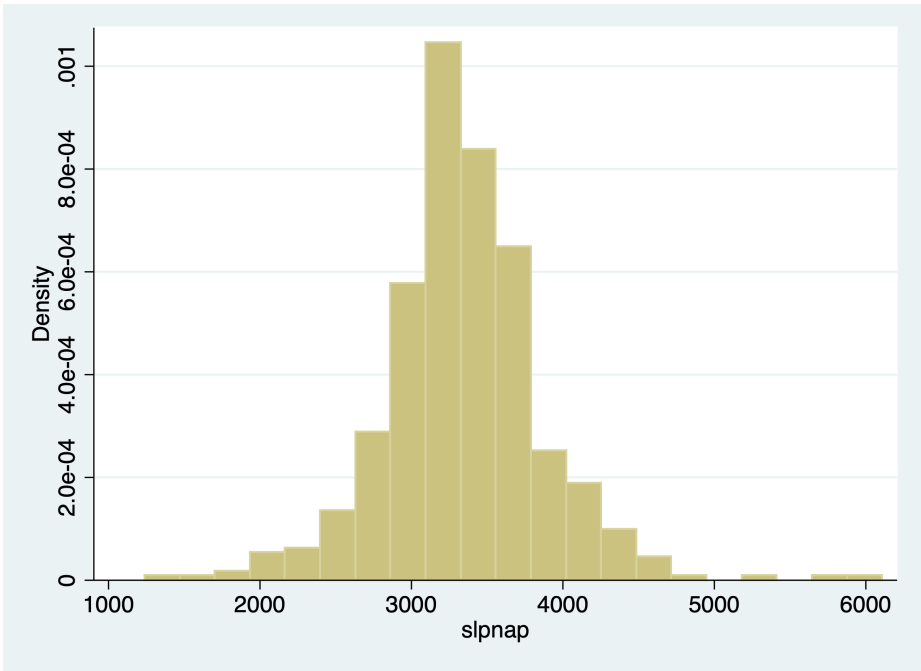


Figure 2.3: Histogram of Minutes Sleeping Per Week

-----+-----					F(5, 233)	=	8.19
Model		14674698.2	5	2934939.64	Prob > F	=	0.0000
Residual		83482611.7	233	358294.471	R-squared	=	0.1495
-----+-----					Adj R-squared	=	0.1313
Total		98157309.9	238	412425.672	Root MSE	=	598.58
-----+-----							
D.slpnap		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+-----							
totwrk							
D1.		-.2266694	.036054	-6.29	0.000	-.2977029	-.1556359
educ							
D1.		-.0244717	48.75938	-0.00	1.000	-96.09008	96.04113
marr							
D1.		104.2139	92.85536	1.12	0.263	-78.72946	287.1574
yngkid							
D1.		94.6654	87.65252	1.08	0.281	-78.02739	267.3582

gdhlth							
D1.		87.57785	76.59913	1.14	0.254	-63.33758	238.4933
_cons		-92.63404	45.8659	-2.02	0.045	-182.9989	-2.269152

Our Pooled OLS model shows one additional hour worked per year is associated with a 0.15 fewer minutes per week of sleep, while our FD model one additional hour worked per year is associated with a 0.23 fewer minutes of sleep per week.

Similar except time dummy and base intercepts look at the difference between educ in Pooled and FE models. There is likely a confounder between ability sleep and education.

```
esttab OLS FD, mtitles se scalars(F r2) drop(0.d81)
```

	(1) OLS	(2) FD
1.d81	-77.03 (47.97)	
totwrk	-0.150*** (0.0244)	
educ	-20.26* (8.000)	
marr	-42.19 (53.08)	
yngkid	74.26 (77.33)	
gdhlth	29.04 (59.58)	
D.totwrk		-0.227*** (0.0361)
D.educ		-0.0245 (48.76)
D.marr		104.2 (92.86)

D.yngkid		94.67 (87.65)
D.gdhlth		87.58 (76.60)
_cons	3958.8*** (121.8)	-92.63* (45.87)

N	478	239
F	8.497	8.191
r2	0.0977	0.150

Standard errors in parentheses

* p<0.05, ** p<0.01, *** p<0.001

Let's look at elasticities

```
gen lnslpnap=ln(slpnap)
gen lntotwrk=ln(totwrk)
gen lneduc=ln(educ)
```

FD Model

```
reg d.lnslpnap d.lntotwrk d.lneduc d.marr d.yngkid d.gdhlth
```

Source	SS	df	MS	Number of obs	=	214
Model	.805249181	5	.161049836	F(5, 208)	=	4.17
Residual	8.03583239	208	.03863381	Prob > F	=	0.0012
-----				R-squared	=	0.0911
Total	8.84108157	213	.041507425	Adj R-squared	=	0.0692
				Root MSE	=	.19655

D.lnslpnap	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	

lntotwrk						
D1.	-.0848256	.0206499	-4.11	0.000	-.1255354	-.0441158
lneduc						
D1.	-.0289507	.2628724	-0.11	0.912	-.5471864	.489285
marr						

D1.		-.0038529	.0323002	-0.12	0.905	-.0675306	.0598247
yngkid							
D1.		.0056761	.0290466	0.20	0.845	-.0515874	.0629397
gdhlth							
D1.		.048008	.0266772	1.80	0.073	-.0045843	.1006004
_cons		-.0204204	.0155593	-1.31	0.191	-.0510947	.0102538

A 1% increase in hours worked per year is associated with a 0.08% decrease in sleep per week.

2.2.2 Distributed Lag of Crime Rates on Clear-up Rate

Lesson: Lagged variables can be important explanatory variables *Note:* We'll get into this a bit more in Wooldridge Chapter 10-12.

We will get into time series later in the course, but we will have a preview. With panel data, we are able to see the same observation over time. Given this, we can see if lagged values affect current values for the cross-sectional unit of observation.

Eide(1994) wants to assess if prior clear-up rates for crime have a relationship with crime rates in the current time period. The dependent variable of interest is the current period crime rate. The variable of interest is the clear-up percentage which is the rate of crimes that have lead to conviction.

We will use a log-log model and use a first-difference.

$$\Delta \ln(\text{crime}_{i,t}) = \delta_0 + \beta_1 \Delta \ln(\text{clearup}_{i,t-1}) + \beta_2 \Delta \ln(\text{clearup}_{i,t-2}) + \Delta \varepsilon_{i,t}$$

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use crime3.dta, clear
reg clcrime cclrprc1 cclrprc2
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

Source		SS	df	MS	Number of obs	=	53
Model		1.42294697	2	.711473484	F(2, 50)	=	5.99
Residual		5.93723904	50	.118744781	Prob > F	=	0.0046
					R-squared	=	0.1933
					Adj R-squared	=	0.1611

Total | 7.36018601 52 .141542039 Root MSE = .34459

clcrime	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
cclrprc1	-.0040475	.0047199	-0.86	0.395	-.0135276	.0054326
cclrprc2	-.0131966	.0051946	-2.54	0.014	-.0236302	-.0027629
_cons	.0856556	.0637825	1.34	0.185	-.0424553	.2137665

Our model tells us that 1 percent increase in getting crimes to conviction is associated with a reduction in the current period crime rate of about 0.013 percent. Assuming this model is specified well, crime rates are sensitive to clear-up rates from the prior 2 years.

2.2.3 Enterprise Zones and Unemployment

Lesson: We need to be aware of heteroskedasticity and serial correlation

We should test for heteroskedasticity and serial correlation when running our First-Difference Estimator.

Serial correlation means that the differences in the idiosyncratic errors between time periods are correlated, which is a violation of our assumption that there is no serial correlation for our first-difference estimator.

We'll use data from Papke (1994) who studied the effect of Indiana's enterprise zone program on unemployment claims. Do enterprise zones increase employment and reduce unemployment claims?

She analyzes 22 cities in Indiana from 1980 to 1988, since 6 enterprise zones were established in 1984. Twelve cities did not receive the enterprise zones, while 10 cities did get enterprise zones.

We'll use a simple analysis that the change in natural log of unemployment claims is a function of time period binaries, change in enterprise zones, and the idiosyncratic error.

One way to do it is to use the data already created.

$$\ln(UnempClaims_{i,t}) = \beta_0 + \beta_1 EnterpriseZone_{i,t} + a_t + a_i + \varepsilon_{i,t}$$

Pooled OLS

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use "ezunem.dta", clear
reg luelms i.d82 i.d83 i.d84 i.d85 i.d86 i.d87 i.d88 cez
```

```
/Users/Sam/Desktop/Econ 645/Data/Wooldridge
```

Source	SS	df	MS	Number of obs	=	176
				F(8, 167)	=	10.39
Model	29.2551235	8	3.65689043	Prob > F	=	0.0000
Residual	58.7753327	167	.3519481	R-squared	=	0.3323
				Adj R-squared	=	0.3003
Total	88.0304561	175	.503031178	Root MSE	=	.59325

luclms	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
1.d82	.4571276	.1788722	2.56	0.011	.1039853	.8102699
1.d83	.1023765	.1788722	0.57	0.568	-.2507658	.4555188
1.d84	-.2821805	.1882109	-1.50	0.136	-.6537598	.0893988
1.d85	-.3150721	.1830816	-1.72	0.087	-.6765248	.0463805
1.d86	-.3470941	.1788722	-1.94	0.054	-.7002364	.0060481
1.d87	-.614778	.1788722	-3.44	0.001	-.9679202	-.2616357
1.d88	-.9534624	.1788722	-5.33	0.000	-1.306605	-.6003201
cez	-.0139924	.2146822	-0.07	0.948	-.4378332	.4098484
_cons	11.37276	.1264818	89.92	0.000	11.12305	11.62247

The Pooled OLS shows that an enterprise zone with unemployment insurance claims. Let's say we didn't have the data already created. What we'll need to do is to set the panel data with `xtset` unit timeperiod Set the Panel Data

```
xtset city year
```

```

panel variable:  city (strongly balanced)
time variable:  year, 1980 to 1988
delta: 1 unit

```

We can use the `d.` operator to use a first difference First Difference

```
reg d.luclms i.d82 i.d83 i.d84 i.d85 i.d86 i.d87 i.d88 d.ez
```

Source	SS	df	MS	Number of obs	=	176
				F(8, 167)	=	34.50
Model	12.8826331	8	1.61032914	Prob > F	=	0.0000
Residual	7.79583815	167	.046681666	R-squared	=	0.6230
				Adj R-squared	=	0.6049
Total	20.6784713	175	.118162693	Root MSE	=	.21606

D.luc1ms	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
1.d82	.7787595	.0651444	11.95	0.000	.6501469	.9073721
1.d83	-.0331192	.0651444	-0.51	0.612	-.1617318	.0954934
1.d84	-.0171382	.0685455	-0.25	0.803	-.1524655	.1181891
1.d85	.323081	.0666774	4.85	0.000	.1914417	.4547202
1.d86	.292154	.0651444	4.48	0.000	.1635413	.4207666
1.d87	.0539481	.0651444	0.83	0.409	-.0746645	.1825607
1.d88	-.0170526	.0651444	-0.26	0.794	-.1456652	.1115601
ez						
D1.	-.1818775	.0781862	-2.33	0.021	-.3362382	-.0275169
_cons	-.3216319	.046064	-6.98	0.000	-.4125748	-.2306891

When we use a FD estimator, a city that adopts an enterprise is associated with a reduction in unemployment insurance claims by $(e^{-0.1819}-1)*100\% = 16.6\%$

Test for Heteroskedasticity:

Bruesch-Pagan/Cameron-Trivedi Test using *estat* post-estimation command

```
estat imtest
```

Cameron & Trivedi's decomposition of IM-test

Source	chi2	df	p
Heteroskedasticity	12.85	9	0.1697
Skewness	13.04	8	0.1104
Kurtosis	0.11	1	0.7414
Total	26.00	18	0.0998

White Test using *estat* postestimation command

```
estat imtest, white
```

White's test for H_0 : homoskedasticity
against H_a : unrestricted heteroskedasticity

```
chi2(9)      =    12.85
Prob > chi2   =    0.1697
```

Cameron & Trivedi's decomposition of IM-test

Source	chi2	df	p
Heteroskedasticity	12.85	9	0.1697
Skewness	13.04	8	0.1104
Kurtosis	0.11	1	0.7414
Total	26.00	18	0.0998

We fail to reject the null hypothesis of homoskedastic standard errors.

Test for autocorrelation We can use the command xtserial to test for serial correlation, since we already set up our panel above.

```
help xtserial
```

```
xtserial luc1ms d83 d84 d85 d86 d87 d88 cez, output
```

```
Linear regression              Number of obs   =    154
                              F(7, 21)         =   102.85
                              Prob > F          =    0.0000
                              R-squared         =    0.4711
                              Root MSE      =    .28063
```

(Std. Err. adjusted for 22 clusters in city)

D.luc1ms	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]
d83					
D1.	-.3547511	.0410541	-8.64	0.000	-.4401277 -.2693745
d84					
D1.	-.7240107	.0816269	-8.87	0.000	-.893763 -.5542583
d85					
D1.	-.7620015	.0540757	-14.09	0.000	-.8744581 -.6495448
d86					

```

D1. | -.8042218 .0490039 -16.41 0.000 -.906131 -.7023126
|
d87 |
D1. | -1.071906 .0496977 -21.57 0.000 -1.175258 -.9685535
|
d88 |
D1. | -1.41059 .065833 -21.43 0.000 -1.547497 -1.273683
|
cez |
D1. | -.070083 .0650804 -1.08 0.294 -.205425 .065259
-----

```

Wooldridge test for autocorrelation in panel data

H0: no first-order autocorrelation

F(1, 21) = 120.981

Prob > F = 0.0000

We reject the null hypothesis that the differences in idiosyncratic errors is 0.

What can we do for this serial correlation? We can cluster our standard errors to account for the serial correlation within units.

Robust for serial correlation within unit of analysis clusters

```
reg d.luc1ms i.d82 i.d83 i.d84 i.d85 i.d86 i.d87 i.d88 d.cez, robust cluster(city)
```

note: 1.d88 omitted because of collinearity

```

Linear regression              Number of obs   =      154
                              F(7, 21)         =      41.20
                              Prob > F          =      0.0000
                              R-squared          =      0.6333
                              Root MSE       =      .21865

```

(Std. Err. adjusted for 22 clusters in city)

```

-----+-----
              |               Robust
D.luc1ms |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-----+-----
    1.d82 |   .7958121   .0577076    13.79   0.000   .6758025   .9158217
    1.d83 |  -.0160666   .0565946    -0.28   0.779  -.1337615   .1016282
    1.d84 |  -.0305751   .0742049    -0.41   0.684  -.1848926   .1237424
    1.d85 |   .3006937   .0568959     5.28   0.000   .1823723   .4190151
    1.d86 |   .2964642   .0786548     3.77   0.001   .1328926   .4600357
    1.d87 |   .0710007   .0462206     1.54   0.139  -.0251204   .1671217
    1.d88 |           0 (omitted)

```

cez							
D1.		-.070083	.0653029	-1.07	0.295	-.2058877	.0657217
_cons		-.3386845	.0359416	-9.42	0.000	-.4134292	-.2639397

Once we account for the serial correlation in the idiosyncratic error, our coefficient on the change in enterprise zone becomes statistically insignificant.

Testing for serial correlation, and Clustering your standard error at the treatment level is an important test and step with FD estimators.

When we get to Fixed Effects (Within) estimator, the assumption is slightly different. For First-Difference Estimator, the DIFFERENCE in idiosyncratic errors cannot be correlated, but for Fixed-Effects (Within) Estimator the assumption states that only the idiosyncratic errors cannot be correlated.

2.2.4 County Crimes in North Carolina

Lesson: First-Difference Estimator (and Fixed Effects Estimator) cannot fix simultaneity bias. We need an valid instrument to deal with simultaneity bias.

Cornwell and Trumbull (1994) analyzed data on 90 counties in North Carolina from 1981 to 1987 to using panel data to account for time-invariant effects. Their cross-sectional unit of observation is the county (not individuals within a county). They want to know what the is the effect of police per capita on crime rates.

For our model the dependent variable is the change in the natural log of crimes per person (*lcrimrte*). They are interested in estimating the sensitivity of police per capita (*polpc*) on crime rate, so they use the difference in natural log of police per capita. This will get you elasticities, which provide useful interpretations of percentage increases.

They also control for the difference in natural log of probability of arrest (*prbarr*), the probability of conviction (*prbconv*), the probability of serving time in prison given a conviction (*prbpris*), and the average sentence length.

$$\ln(\text{crimrate}_{i,t}) = \beta_0 + \beta_1 \ln(\text{prbcon}_{i,t}) + \beta_2 \ln(\text{prbpris}_{i,t}) + \beta_3 \ln(\text{avgsen}_{i,t}) + \beta_4 \ln(\text{polpc}_{i,t}) + a_t + a_i + \varepsilon_{i,t}$$

Pooled OLS

```
cd "/Users/Sam/Desktop/Econ 645/Data/Wooldridge"
use "crime4.dta", clear
reg lcrmrte i.d82 i.d83 i.d84 i.d85 i.d86 i.d87 lprbarr lprbconv lprbpris lavgsen lpolpc
```

32 CHAPTER 2. PANEL DATA PART 1 - POOLED DATA AND FIRST DIFFERENCE

/Users/Sam/Desktop/Econ 645/Data/Wooldridge

Source	SS	df	MS	Number of obs	=	630
				F(11, 618)	=	74.49
Model	117.644669	11	10.6949699	Prob > F	=	0.0000
Residual	88.735673	618	.143585231	R-squared	=	0.5700
				Adj R-squared	=	0.5624
Total	206.380342	629	.328108652	Root MSE	=	.37893

lcrmrte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
1.d82	.0051371	.057931	0.09	0.929	-.1086284	.1189026
1.d83	-.043503	.0576243	-0.75	0.451	-.1566662	.0696601
1.d84	-.1087542	.057923	-1.88	0.061	-.222504	.0049957
1.d85	-.0780454	.0583244	-1.34	0.181	-.1925835	.0364928
1.d86	-.0420791	.0578218	-0.73	0.467	-.15563	.0714719
1.d87	-.0270426	.056899	-0.48	0.635	-.1387815	.0846963
lprbarr	-.7195033	.0367657	-19.57	0.000	-.7917042	-.6473024
lprbconv	-.5456589	.0263683	-20.69	0.000	-.5974413	-.4938765
lprbpris	.2475521	.0672268	3.68	0.000	.1155314	.3795728
lavgsen	-.0867575	.0579205	-1.50	0.135	-.2005023	.0269872
lpolpc	.3659886	.0300252	12.19	0.000	.3070248	.4249525
_cons	-2.082293	.2516253	-8.28	0.000	-2.576438	-1.588149

First Difference

```
xtset county year
```

```

panel variable:  county (strongly balanced)
time variable:   year, 81 to 87
delta:           1 unit

```

With xtset, we can use the d. for our explanatory variables.

First-Difference

```
reg d.lcrmrte i.d83 i.d84 i.d85 i.d86 i.d87 d.lprbarr d.lprbconv d.lprbpris d.lavgsen
```

Source	SS	df	MS	Number of obs	=	540
				F(10, 529)	=	40.32
Model	9.60042828	10	.960042828	Prob > F	=	0.0000
Residual	12.5963755	529	.023811674	R-squared	=	0.4325
				Adj R-squared	=	0.4218

Total | 22.1968038 539 .041181454 Root MSE = .15431

D.lcrmrte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
1.d83	-.0998658	.0238953	-4.18	0.000	-.1468071	-.0529246
1.d84	-.0479374	.0235021	-2.04	0.042	-.0941063	-.0017686
1.d85	-.0046111	.0234998	-0.20	0.845	-.0507756	.0415533
1.d86	.0275143	.0241494	1.14	0.255	-.0199261	.0749548
1.d87	.0408267	.0244153	1.67	0.095	-.0071361	.0887895
lprbarr						
D1.	-.3274942	.0299801	-10.92	0.000	-.3863889	-.2685994
lprbconv						
D1.	-.2381066	.0182341	-13.06	0.000	-.2739268	-.2022864
lprbpris						
D1.	-.1650462	.025969	-6.36	0.000	-.2160613	-.1140312
lavgsen						
D1.	-.0217607	.0220909	-0.99	0.325	-.0651574	.021636
lpolpc						
D1.	.3984264	.026882	14.82	0.000	.3456177	.451235
_cons	.0077134	.0170579	0.45	0.651	-.0257961	.0412229

So our model is estimating that a 1% increase in police per capita increases the crime rate by about 0.4%. We would expect that increases in police per capita would reduce crime rates. This is likely shows that there are problems in our model.

What is happening? Likely simultaneity bias is occurring. Areas with high crime rates have more police per capita, which more police per capita is associated with higher crime rates. It's a circular/endogenous reference, so we will need an valid instrument to deal with this simultaneity bias/endogeneity.

We can test for heteroskedasticity and serial correlation, but this only affects our standard errors, not our coefficients.

Test for autocorrelation with one lag AR(1). We can see that serial correlation is a problem, which means we should cluster our standard errors at the county level.

```

predict r, residual
gen lag_r =l.r
reg r lag_r i.d83 i.d84 i.d85 i.d86 i.d87 d.lprbarr d.lprbconv d.lprbpris d.lavgsen d.

```

(90 missing values generated)

(180 missing values generated)

note: 1.d87 omitted because of collinearity

Source	SS	df	MS	Number of obs	=	450
				F(10, 439)	=	2.35
Model	.564663971	10	.056466397	Prob > F	=	0.0102
Residual	10.528838	439	.023983686	R-squared	=	0.0509
				Adj R-squared	=	0.0293
Total	11.093502	449	.024707131	Root MSE	=	.15487

r	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
lag_r	-.2332117	.0488802	-4.77	0.000	-.32928	-.1371435
1.d83	-.0012379	.0234019	-0.05	0.958	-.0472317	.0447558
1.d84	-.0016543	.0236303	-0.07	0.944	-.0480968	.0447883
1.d85	-.0006044	.0236114	-0.03	0.980	-.0470098	.0458009
1.d86	.0008841	.0233216	0.04	0.970	-.0449517	.0467199
1.d87	0	(omitted)				
lprbarr						
D1.	.0082875	.0330698	0.25	0.802	-.0567074	.0732823
lprbconv						
D1.	-.0036161	.0200155	-0.18	0.857	-.0429542	.0357221
lprbpris						
D1.	.0017131	.027855	0.06	0.951	-.0530326	.0564588
lavgsen						
D1.	-.0125831	.0245006	-0.51	0.608	-.0607362	.03557
lpolpc						
D1.	.0214883	.0282688	0.76	0.448	-.0340706	.0770473
_cons	.0005347	.0167411	0.03	0.975	-.032368	.0334373

We can always use our xtserial command to test for AR(1) serial correlation.

```
reg d.lcrmrte i.d83 i.d84 i.d85 i.d86 i.d87 d.lprbarr d.lprbconv d.lprbpris d.lavgsen d.lpolpc
xtserial lcrmrte d83 d84 d85 d86 d87 lprbarr lprbconv lprbpris lavgsen lpolpc, output
```

Source	SS	df	MS	Number of obs	=	540
				F(10, 529)	=	40.32
Model	9.60042828	10	.960042828	Prob > F	=	0.0000
Residual	12.5963755	529	.023811674	R-squared	=	0.4325
				Adj R-squared	=	0.4218
Total	22.1968038	539	.041181454	Root MSE	=	.15431

D.lcrmrte	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
1.d83	-.0998658	.0238953	-4.18	0.000	-.1468071	-.0529246
1.d84	-.0479374	.0235021	-2.04	0.042	-.0941063	-.0017686
1.d85	-.0046111	.0234998	-0.20	0.845	-.0507756	.0415533
1.d86	.0275143	.0241494	1.14	0.255	-.0199261	.0749548
1.d87	.0408267	.0244153	1.67	0.095	-.0071361	.0887895
lprbarr						
D1.	-.3274942	.0299801	-10.92	0.000	-.3863889	-.2685994
lprbconv						
D1.	-.2381066	.0182341	-13.06	0.000	-.2739268	-.2022864
lprbpris						
D1.	-.1650462	.025969	-6.36	0.000	-.2160613	-.1140312
lavgsen						
D1.	-.0217607	.0220909	-0.99	0.325	-.0651574	.021636
lpolpc						
D1.	.3984264	.026882	14.82	0.000	.3456177	.451235
_cons	.0077134	.0170579	0.45	0.651	-.0257961	.0412229

Linear regression

Number of obs	=	540
F(10, 89)	=	13.68
Prob > F	=	0.0000
R-squared	=	0.4323
Root MSE	=	.15419

(Std. Err. adjusted for 90 clusters in county)							
D.lcrmrte	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]		
d83							
D1.	-.0920147	.0146354	-6.29	0.000	-.121095	-.0629345	
d84							
D1.	-.1323557	.0179759	-7.36	0.000	-.1680734	-.096638	
d85							
D1.	-.1293278	.0232716	-5.56	0.000	-.1755679	-.0830876	
d86							
D1.	-.0938692	.0209561	-4.48	0.000	-.1355086	-.0522298	
d87							
D1.	-.0449434	.0236533	-1.90	0.061	-.091942	.0020552	
lprbarr							
D1.	-.3263268	.0561808	-5.81	0.000	-.437957	-.2146967	
lprbconv							
D1.	-.2375214	.0392585	-6.05	0.000	-.3155272	-.1595155	
lprbpris							
D1.	-.1645109	.0457092	-3.60	0.001	-.255334	-.0736878	
lavgsen							
D1.	-.0246909	.0252756	-0.98	0.331	-.0749129	.0255312	
lpolpc							
D1.	.3976861	.1024878	3.88	0.000	.1940451	.6013271	

Wooldridge test for autocorrelation in panel data

H0: no first-order autocorrelation

F(1, 89) = 22.065

Prob > F = 0.0000

We can also test for heteroskedasticity

```
quietly reg d.lcrmte i.d83 i.d84 i.d85 i.d86 i.d87 d.lprbarr d.lprbconv d.lprbpris d.lavgsen d.l
rvfplot, yline(0)
graph export "/Users/Sam/Desktop/Econ 645/Stata/week3_htsk.png", replace
```

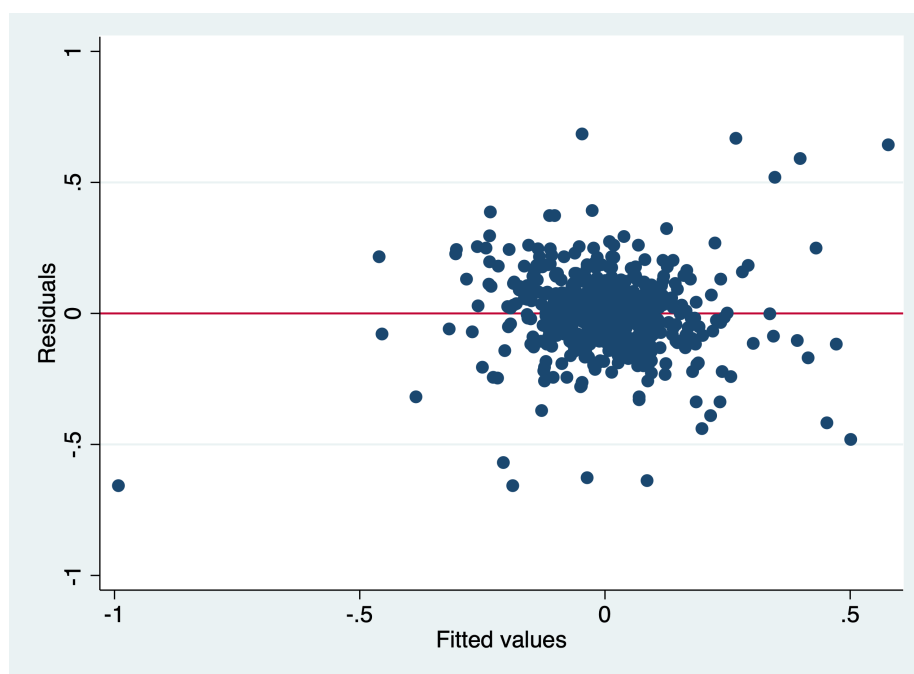


Figure 2.4: Plotting Errors

We can use the White Test and Breusch-Pagan, which yield different results

```
estat imtest, white
estat hettest
```

White's test for H_0 : homoskedasticity
against H_a : unrestricted heteroskedasticity

```
chi2(50)      =    257.57
Prob > chi2   =    0.0000
```

Cameron & Trivedi's decomposition of IM-test

Source	chi2	df	p
-----+-----			

38 CHAPTER 2. PANEL DATA PART 1 - POOLED DATA AND FIRST DIFFERENCE

Heteroskedasticity		257.57	50	0.0000
Skewness		62.96	10	0.0000
Kurtosis		12.82	1	0.0003
-----+				
Total		333.35	61	0.0000

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity

Ho: Constant variance

Variables: fitted values of D.lcrmte

chi2(1) = 1.16

Prob > chi2 = 0.2807

Re-estimate with Robust Standard Errors clustered at the county level and cluster by county (deal with error correlations within counties).

`reg d.lcrmte i.d83 i.d84 i.d85 i.d86 i.d87 d.lprbarr d.lprbconv d.lprbpris d.lavgsen`

Linear regression	Number of obs	=	540
	F(10, 89)	=	13.56
	Prob > F	=	0.0000
	R-squared	=	0.4325
	Root MSE	=	.15431

(Std. Err. adjusted for 90 clusters in county)

		Robust					
D.lcrmte		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+							
1.d83		-.0998658	.0222563	-4.49	0.000	-.1440887	-.055643
1.d84		-.0479374	.0200531	-2.39	0.019	-.0877825	-.0080923
1.d85		-.0046111	.02503	-0.18	0.854	-.0543453	.045123
1.d86		.0275143	.0211829	1.30	0.197	-.0145756	.0696043
1.d87		.0408267	.0241102	1.69	0.094	-.0070797	.0887331
lprbarr							
D1.		-.3274942	.0564281	-5.80	0.000	-.4396157	-.2153727
lprbconv							
D1.		-.2381066	.0395843	-6.02	0.000	-.3167598	-.1594534
lprbpris							
D1.		-.1650462	.0457923	-3.60	0.001	-.2560345	-.074058

lavg	sen						
D1.			-.0217607	.02582	-0.84	0.402	-.0730644
lpol	pc						
D1.			.3984264	.1029342	3.87	0.000	.1938983
_cons			.0077134	.0137846	0.56	0.577	-.0196763

Even accounting for heteroskedasticity and serial correlation, our model is likely biased by simultaneity bias.

Chapter 3

Labeling Data

Mitchell's 5th chapter has a bunch of useful information, but the most important parts are *label define* (and its two options of *add* and *modify*), *label values*, and time formatting.

3.1 Describing datasets

We have seen the *describe* command before, but it is a very useful command to being working with data. It provides the variable name, storage type, display format, value label, and variable label, Let's get some data on the survey of graduate students.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"  
use "survey7.dta", clear  
describe
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```
(Survey of graduate students)
```

```
Contains data from survey7.dta
```

obs:	8	Survey of graduate students
vars:	11	5 May 2020 14:37
size:	400	(_dta has notes)

	storage	display	value	
variable name	type	format	label	variable label

```

-----
id          float   %9.0g          Unique identification variable
STUDENTVARS float   %9.0g          STUDENT VARIABLES =====
gender      float   %9.0g          mf          Gender of student
race        float   %19.0g         racelab   * Race of student
bday        float   %td..          Date of birth of student
income      float   %11.1fc        Income of student
havechild   float   %18.0g         havelab   * Given birth to a child?
KIDVARS     float   %9.0g          KID VARIABLES =====
kidname     str10   %-10s          Name of child
ksex        float   %15.0g         mfkid     * Sex of child
kday        float   %td..          Date of birth of child
                                     * indicated variables have notes
-----

```

Sorted by:

We also have a short option, but it just contain general information.

```
describe, short
```

```

Contains data from survey7.dta
  obs:          8          Survey of graduate students
  vars:         11          5 May 2020 14:37
  size:         400
Sorted by:

```

We can subset the variables we want to describe if we want

```
describe id gender race
```

```

-----
          storage  display  value
variable name  type   format  label    variable label
-----
id            float   %9.0g          Unique identification variable
gender        float   %9.0g          mf          Gender of student
race          float   %19.0g         racelab   * Race of student

```

Finally, the command *codebook* provides a deep dive into your dataset. This is very useful for looking at the value labels. We only see the value label name in the *describe* command, but the *codebook* command provides more information, such as type of variable, label name, range of values, unique values, missing, value labels, missing value labels (if any).

codebook

id Unique identification variable

```

      type:  numeric (float)
      range:  [1,8]
unique values: 8
      units:  1
      missing .:  0/8

```

```

tabulation:  Freq.  Value
              1    1
              1    2
              1    3
              1    4
              1    5
              1    6
              1    7
              1    8

```

STUDENTVARS STUDENT VARIABLES =====

```

      type:  numeric (float)
      range:  [.,.]
unique values: 0
      units:  .
      missing .:  8/8

```

```

tabulation:  Freq.  Value
              8    .

```

gender Gender of student

```

      type:  numeric (float)
      label:  mf
      range:  [1,2]
unique values: 2
      units:  1
      missing .:  0/8

```

```

tabulation:  Freq.  Numeric  Label
              3         1   Male
              5         2  Female

```

race	Race of
------	---------

```

      type: numeric (float)
      label: racelab

      range: [1,5]
unique values: 5
      units: 1
missing .: 0/8

      tabulation: Freq.  Numeric  Label
                  2         1  White
                  2         2  Asian
                  2         3  Hispanic
                  1         4  African American
                  1         5  Other

```

bday	Date of birth of
------	------------------

```

      type: numeric daily date (float)

      range: [389,7935]
or equivalently: [24jan1961,22sep1981]
unique values: 8
      units: 1
      units: days
missing .: 0/8

      tabulation: Freq.  Value
                  1  389   24jan1961
                  1 3027   15apr1968
                  1 4160   23may1971
                  1 4924   25jun1973
                  1 5036   15oct1973
                  1 6059   03aug1976
                  1 6391   01jul1977
                  1 7935   22sep1981

```

income	Income of
--------	-----------

```

      type: numeric (float)

      range: [545.23,1284354.5]
unique values: 8
      units: .01
missing .: 0/8

```

```

tabulation:  Freq.  Value
              1  545.22998
              1  4500.9199
              1  10500.93
              1  45234.129
              1  109452.11
              1  120102.32
              1  124313.45
              1  1284354.5

```

```

-----
havechild                                         Given birth to a child?
-----

```

```

              type:  numeric (float)
              label:  havelab

              range:  [0,1]                      units:  1
unique values:  2                      missing .:  0/8
unique mv codes: 1                      missing .*:  3/8

```

```

tabulation:  Freq.  Numeric  Label
              1      0  Dont Have Child
              4      1  Have Child
              3      .n  NA

```

```

-----
KIDVARS                                         KID VARIABLES =====
-----

```

```

              type:  numeric (float)

              range:  [.,.]                      units:  .
unique values:  0                      missing .:  8/8

```

```

tabulation:  Freq.  Value
              8      .

```

```

-----
kidname                                         Name of child
-----

```

```

              type:  string (str10), but longest is str9

unique values:  5                      missing "":  0/8

```

```

tabulation:  Freq.  Value
              4      ""
              1  "Catherine"
              1  "Robin"
              1  "Sally"
              1  "Samuell"

```

```

warning:  variable has leading and trailing blanks

```

```

-----
ksex
-----

```

```

-----
Sex
-----

```

```

      type:  numeric (float)
      label:  mfkid

      range:  [1,2]                      units:  1
unique values: 2                      missing .:  0/8
unique mv codes: 2                    missing .*:  5/8

```

```

tabulation:  Freq.  Numeric  Label
              1         1   Male
              2         2  Female
              4         .n   NA
              1         .u  Unknown

```

```

-----
kday
-----

```

```

-----
Date of birth
-----

```

```

      type:  numeric daily date (float)

      range:  [12888,15932]              units:  1
or equivalently: [15apr1995,15aug2003]  units:  days
unique values:  4                      missing .:  4/8

```

```

tabulation:  Freq.  Value
              1  12888  15apr1995
              1  14019  20may1998
              1  14256  12jan1999
              1  15932  15aug2003
              4      .

```

We can go by variables.

```
codebook race
```

```

race                                                    Race of student
-----
                                type:  numeric (float)
                                label:  racelab

                                range:  [1,5]                units:  1
unique values:  5                                missing .:  0/8

tabulation:  Freq.    Numeric  Label
              2         1  White
              2         2  Asian
              2         3  Hispanic
              1         4  African American
              1         5  Other

```

We can go by variables and notes

```
codebook havechild, notes
```

```

havechild                                                    Given birth to a child?
-----
                                type:  numeric (float)
                                label:  havelab

                                range:  [0,1]                units:  1
unique values:  2                                missing .:  0/8
unique mv codes: 1                                missing .*:  3/8

tabulation:  Freq.    Numeric  Label
              1         0  Dont Have Child
              4         1  Have Child
              3         .n  NA

```

havechild:

1. This variable measures whether a woman has given birth to a child, not just whether she is a parent.
2. The .n (NA) missing code is used for males, because they cannot bear children.
3. The .u (Unknown) missing code for a female indicating it is unknown if she has a child.

We can look at the variable and missing value labels with the option *mv*. I recommend that you don't label the missing values unless it is absolutely nec-

essary. Different types of missing values besides “.” cause problems down the road, especially with the *marginsplot* command.

```
codebook ksex, mv
```

```
ksex
```

```
Sex o
```

```

                                type:  numeric (float)
                                label:  mfkid

                                range:  [1,2]                      units:  1
                                unique values:  2                  missing .:  0/8
                                unique mv codes: 2                  missing .*:  5/8

                                tabulation:  Freq.   Numeric  Label
                                                1           1  Male
                                                2           2  Female
                                                4           .n  NA
                                                1           .u  Unknown

                                missing values:  havechild==mv --> ksex==mv
                                                kbday==mv --> ksex==mv
```

If you are interested in the different languages labels it is on page 112.

The *lookfor* command will return all variables with the search word. This is a bit redundant, since this is available in the variable window. But, it provides more space to look.

```
lookfor birth
```

variable name	storage type	display format	value label	variable label
bday	float	%td..		Date of birth of student
havechild	float	%18.0g	havelab *	Given birth to a child?
kbday	float	%td..		Date of birth of child

We can also search for the notes by the search word

```
notes search birth
```

```
havechild:
```

1. This variable measures whether a woman has given birth to a child, not just whether she is a parent.

We can see the formats of the variables as well

```
list income bday
describe income bday
```

```

      |      income      bday |
      |-----|
1. |    10,500.9    01/24/61 |
2. |    45,234.1    04/15/68 |
3. |  1,284,354.5    05/23/71 |
4. |   124,313.5    06/25/73 |
5. |   120,102.3    09/22/81 |
      |-----|
6. |      545.2    10/15/73 |
7. |   109,452.1    07/01/77 |
8. |    4,500.9    08/03/76 |
+-----+
```

	storage	display	value	
variable name	type	format	label	variable label
income	float	%11.1fc		Income of student
bday	float	%td..		Date of birth of student

We can see that the format for income is %11.1fc and the format for bday is %td

3.2 Labeling variables

Labeling the variables is a very helpful shortcut to describe what the variable contain without having to go back to the data dictionary. Sometimes we want a short and concise label if we are exporting labels to regression tables, or sometimes we want longer variable labels to give us context of the variable.

Let's get some data on graduate students and use the *describe* command.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use "survey1.dta", clear
describe
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

Contains data from survey1.dta

```
obs:      8
vars:     9                      1 Jan 2010 12:13
size:    432
```

variable name	storage type	display format	value label	variable label
id	float	%9.0g		
gender	float	%9.0g		
race	float	%9.0g		
havechild	float	%9.0g		
ksex	float	%9.0g		
bdays	str10	%10s		
income	float	%9.0g		
kbdays	str10	%10s		
kidname	str10	%10s		

Sorted by:

We have no variable labels, so we will need to provide some so future users have an understand what the data are. We will use the *label variable* command to describe the variable.

```
label variable id "Identification variable"
label variable gender "Gender of the student"
label variable race "Race of the student"
label variable havechild "Given birth to a child"
label variable ksex "Sex of child"
label variable bday "Birthday of student"
label variable income "Income of student"
label variable kbdays "Birthday of child"
label variable kidname "Name of child"
describe
```

Contains data from survey1.dta

```
obs:      8
vars:     9                      1 Jan 2010 12:13
size:    432
```

variable name	storage type	display format	value label	variable label
id	float	%9.0g		Identification variable

gender	float	%9.0g	Gender of the student
race	float	%9.0g	Race of the student
havechild	float	%9.0g	Given birth to a child
ksex	float	%9.0g	Sex of child
bdays	str10	%10s	Birthday of student
income	float	%9.0g	Income of student
kbdays	str10	%10s	Birthday of child
kidname	str10	%10s	Name of child

Sorted by:

We can simply change the variable label with running the command again with the new variable label.

```
label variable id "Unique identification variable"
describe
save survey2, replace
```

Contains data from survey1.dta

```
obs:      8
vars:      9
size:     432
1 Jan 2010 12:13
```

variable name	storage type	display format	value label	variable label
id	float	%9.0g		Unique identification variable
gender	float	%9.0g		Gender of the student
race	float	%9.0g		Race of the student
havechild	float	%9.0g		Given birth to a child
ksex	float	%9.0g		Sex of child
bdays	str10	%10s		Birthday of student
income	float	%9.0g		Income of student
kbdays	str10	%10s		Birthday of child
kidname	str10	%10s		Name of child

Sorted by:

file survey2.dta saved

3.3 Labeling values

Labeling values is a very practice way of analyzing data without having to go back to the data dictionary. Labeling values requires two commands:

1. *label define* to define a label that can be used across many different variables.
2. *label variable* to place a label onto a variable.

Labeling values is a bit different than labeling variables, since we need to modify or replace after a label has been defined. If you try to change a label, then you will get an error unless you use *add*, *modify* or *replace*

Let's look at our codebook

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use survey2, clear
codebook
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```
-----
id                                                                    Unique identification v
-----
```

```

                                type:  numeric (float)
                                range:  [1,8]
                                unique values:  8
                                units:  1
                                missing .:  0/8
```

```

tabulation:  Freq.  Value
              1    1
              1    2
              1    3
              1    4
              1    5
              1    6
              1    7
              1    8
```

```
-----
gender                                                                Gender of the
-----
```

```

                                type:  numeric (float)
                                range:  [1,2]
                                unique values:  2
                                units:  1
                                missing .:  0/8
```

```

tabulation:  Freq.  Value
              3    1
              5    2

```

```

race                                                    Race of the student

```

```

type:  numeric (float)

range:  [1,5]                units:  1
unique values:  5            missing .:  0/8

```

```

tabulation:  Freq.  Value
              2    1
              2    2
              2    3
              1    4
              1    5

```

```

havechild                                              Given birth to a child

```

```

type:  numeric (float)

range:  [0,1]                units:  1
unique values:  2            missing .:  0/8
unique mv codes:  1          missing .*:  3/8

```

```

tabulation:  Freq.  Value
              1    0
              4    1
              3   .n

```

```

ksex                                                    Sex of child

```

```

type:  numeric (float)

range:  [1,2]                units:  1
unique values:  2            missing .:  0/8
unique mv codes:  2          missing .*:  5/8

```

```

tabulation:  Freq.  Value

```

```

1 1
2 2
4 .n
1 .u

```

bdays

Birthday of

```

type: string (str10)

unique values: 8                                missing "": 0/8

tabulation: Freq. Value
              1 "1/24/1961"
              1 "10/15/1973"
              1 "4/15/1968"
              1 "5/23/1971"
              1 "6/25/1973"
              1 "7/1/1977"
              1 "8/3/1976"
              1 "9/22/1981"

```

income

Income of

```

type: numeric (float)

range: [545.23,1284354.5]          units: .01
unique values: 8                    missing .: 0/8

tabulation: Freq. Value
              1 545.22998
              1 4500.9199
              1 10500.93
              1 45234.129
              1 109452.11
              1 120102.32
              1 124313.45
              1 1284354.5

```

kdays

Birthday of

```

type: string (str10), but longest is str9

unique values: 5                                missing "": 0/8

tabulation: Freq. Value
              4  ""
              1  "1/12/1999"
              1  "4/15/1995"
              1  "5/20/1998"
              1  "8/15/2003"

warning: variable has leading and trailing blanks

```

```

-----
kidname                                     Name of child
-----

```

```

type: string (str10), but longest is str9

unique values: 5                                missing "": 0/8

tabulation: Freq. Value
              4  ""
              1  "Catherine"
              1  "Robin"
              1  "Sally"
              1  "Samuell"

warning: variable has leading and trailing blanks

```

We have our variable labels from 5.3, but now we need to label the values so replicators can know what the data are without having to reference the data dictionary for every variable. We will use the *label define* command to create a new label.

First we need to define a label with *label define*

```
label define racelabel 1 "White" 2 "Asian" 3 "Hispanic" 4 "Black"
```

Next we need to label the values of the variable with *label values* command and we'll look at the codebook again.

```
label values race racelabel
codebook race
```

```
race
```

```
Race of the student
```

```

      type: numeric (float)
      label: racelabel, but 1 nonmissing value is not labeled

      range: [1,5]                      units: 1
unique values: 5                      missing .: 0/8

      tabulation: Freq.   Numeric   Label
                  2         1   White
                  2         2   Asian
                  2         3   Hispanic
                  1         4   Black
                  1         5

```

We are still missing a value label for 5, which is Other, so we need to modify our defined label `racel`. If we do not modify our label, we will get an error if we try to label values again. We can use the *add* option in `label define`.

```
label define racelabel 5 "Other", add
```

If we want to modify an existing label, we can use the *modify* option in `label define`.

```
label define racelabel 4 "African American", modify
codebook race
```

```

race                                                                    Race of the

```

```

      type: numeric (float)
      label: racelabel

      range: [1,5]                      units: 1
unique values: 5                      missing .: 0/8

      tabulation: Freq.   Numeric   Label
                  2         1   White
                  2         2   Asian
                  2         3   Hispanic
                  1         4   African American
                  1         5   Other

```

Labeling missing is something that I don't recommend, but we'll show an example here.


```

label define mfkid 1 "Male" 2 "Female" .u "Unknown" .n "NA"
label values ksex mfkid
codebook ksex

label define havechildlabel 0 "Don't have a child" 1 "Have a child" .u "Unknown" .n "NA"
label values havechild havechildlabel
codebook havechild

```

ksex Sex of child

```

          type: numeric (float)
          label: mfkid

          range: [1,2]                      units: 1
    unique values: 2                      missing .: 0/8
    unique mv codes: 2                    missing .*: 5/8

```

tabulation:	Freq.	Numeric	Label
	1	1	Male
	2	2	Female
	4	.n	NA
	1	.u	Unknown

havechild Given birth to a child

```

          type: numeric (float)
          label: havechildlabel

          range: [0,1]                      units: 1
    unique values: 2                      missing .: 0/8
    unique mv codes: 1                    missing .*: 3/8

```

tabulation:	Freq.	Numeric	Label
	1	0	Don't have a child
	4	1	Have a child
	3	.n	NA

We can look at our label list to see what we have define so far. We can use the *label list* command to see our labels.

```
label list
```

```
havechildlabel:
    0 Don't have a child
    1 Have a child
.n NA
.u Unknown
mfkid:
    1 Male
    2 Female
.n NA
.u Unknown
racelabel:
    1 White
    2 Asian
    3 Hispanic
    4 African American
    5 Other
```

The *numlabel* command is an interesting command. It takes the guess work out of knowing the numeric value of the category by appending the numeric value with the label value.

```
numlabel racelabel, add
label list racelabel
tabulate race
```

```
racelabel:
    1 1. White
    2 2. Asian
    3 3. Hispanic
    4 4. African American
    5 5. Other
```

Race of the student	Freq.	Percent	Cum.
1	2	25.00	25.00
2	2	25.00	50.00
3	2	25.00	75.00
4	1	12.50	87.50
5	1	12.50	100.00
Total	8	100.00	

And, if we don't like it or don't need it any more, we can remove the numeric values with the *remove* option.

```
numlabel racelabel, remove
tabulate race
```

Race of the student		Freq.	Percent	Cum.
1		2	25.00	25.00
2		2	25.00	50.00
3		2	25.00	75.00
4		1	12.50	87.50
5		1	12.50	100.00
Total		8	100.00	

We can add additional characters with *numlabel* as well, such as “#=” or “#)” with the *mask* option

```
numlabel racelabel, add mask("#) ")
tabulate race
```

Race of the student		Freq.	Percent	Cum.
1		2	25.00	25.00
2		2	25.00	50.00
3		2	25.00	75.00
4		1	12.50	87.50
5		1	12.50	100.00
Total		8	100.00	

We can remove the mask with *remove* plus the *mask* option

```
numlabel racelabel, remove mask("#) ")
tabulate race
```

Race of the student		Freq.	Percent	Cum.
1		2	25.00	25.00
2		2	25.00	50.00

3	2	25.00	75.00
4	1	12.50	87.50
5	1	12.50	100.00
Total	8	100.00	

```
save survey3, replace
```

3.4 Labeling utilities

Stata has label utilities to manage the labels defined. The first one is *label dir* to see the labels names available in a quick and more concise way than using the *codebook* command.

For me, I think that *label list* will be your most useful command here.

Quick check of your label directory

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use survey3, clear
label dir
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```
mfkid
havechildlabel
racelabel
```

The *label list* command gives a more comprehensive view of your labels that includes the value labels associated with the value label name

```
label list
```

```
mfkid:
      1 Male
      2 Female
      .n NA
      .u Unknown
havechildlabel:
      0 Don't have a child
      1 Have a child
      .n NA
```

```

        .u Unknown
racelabel:
    1 White
    2 Asian
    3 Hispanic
    4 African American
    5 Other

```

The *label save* command will save your labels into a do file for future use. Our do file name is stated after the using statement.

```
label save havechildlabel racelabel using surveylabs, replace
type surveylabs.do
```

```

label define havechildlabel 0 `Don't have a child', modify
label define havechildlabel 1 `Have a child', modify
label define havechildlabel .n `NA', modify
label define havechildlabel .u `Unknown', modify
label define racelabel 1 `White', modify
label define racelabel 2 `Asian', modify
label define racelabel 3 `Hispanic', modify
label define racelabel 4 `African American', modify
label define racelabel 5 `Other', modify

```

The *labelbook* command provides information similar to *codebook* but only for the labels that are defined.

```
labelbook
labelbook racelabel
```

```
value label havechildlabel
```

```

-----
values                                labels
range:  [0,1]                        string length:  [2,18]
      N:   4                          unique at full length:  yes
gaps:   no                          unique at length 12:  yes
missing .*: 2                        null string:  no
                                   leading/trailing blanks:  no
                                   numeric -> numeric:  no

definition
    0   Don't have a child
    1   Have a child
    .n  NA

```

```

        .u    Unknown

variables:  havechild

```

```
value label mfkid
```

values		labels
range:	[1,2]	string length: [2,7]
N:	4	unique at full length: yes
gaps:	no	unique at length 12: yes
missing .*:	2	null string: no
		leading/trailing blanks: no
		numeric -> numeric: no

definition

1	Male
2	Female
.n	NA
.u	Unknown

variables: ksex

```
value label racelabel
```

values		labels
range:	[1,5]	string length: [5,16]
N:	5	unique at full length: yes
gaps:	no	unique at length 12: yes
missing .*:	0	null string: no
		leading/trailing blanks: no
		numeric -> numeric: no

definition

1	White
2	Asian
3	Hispanic
4	African American
5	Other

variables: race

```
value label racelabel
```

```

      values                                labels
      range:  [1,5]                        string length:  [5,16]
          N:   5                          unique at full length: yes
      gaps:   no                          unique at length 12: yes
missing .*:  0                          null string:      no
                                         leading/trailing blanks: no
                                         numeric -> numeric: no

definition
  1  White
  2  Asian
  3  Hispanic
  4  African American
  5  Other

variables:  race
```

The *problem* option for *labelbook* provides information to alert the users of any problems

```
labelbook, problem
```

```
no potential problems in dataset survey3.dta
```

We can have a more detailed look with the *detail* and *problem* options

```
labelbook racelabel, problem detail
```

```
value label racelabel
```

```

      values                                labels
      range:  [1,5]                        string length:  [5,16]
          N:   5                          unique at full length: yes
      gaps:   no                          unique at length 12: yes
missing .*:  0                          null string:      no
                                         leading/trailing blanks: no
                                         numeric -> numeric: no

definition
  1  White
```

```

2    Asian
3    Hispanic
4    African American
5    Other

variables:  race

```

no potential problems in dataset survey3.dta

3.5 Labeling variables and values in different languages

We will not be covering this, but if you are interested, please review pages 127-132.

3.6 Using Notes

The *note* command can be helpful for future users or for replicators. If you use the *note* command without specifying the variable, then it is a general note that will show up under the `__dta` note. If you add a variable in front of the *note* command, like `note var1:`, then you will add a note to the variable

Let's add some general notes.

```
note: This was based on the dataset called survey1.txt
```

Adding *TS* to the end adds a timestamp, which is a nice feature.

```
note: The missing values for havechild and chldage were coded using -1 and -2 but were
```

Let's call our notes with the *notes* command.

```
notes
```

```
__dta:
```

1. This was based on the dataset called survey1.txt
2. The missing values for havechild and chldage were coded using -1 and -2 but were converted to .n and .u 24 Sep 2025 17:34

Let's add some notes to our variables


```
note race: The other category includes people who specified multiple races
note race: This is another note
note race: This is a third note
```

We can just call a particular variable for notes

```
notes race
```

We can just call a particular variable for notes

```
race:
  1. The other category includes people who specified multiple races
  2. This is another note
  3. This is a third note
```

Let say we added an unhelpful note, then we can drop it with the *notes drop* command and we want to drop the second note.

```
notes drop race in 2
notes race
```

```
(1 note dropped)
```

```
race:
  1. The other category includes people who specified multiple races
  3. This is a third note
```

Notice that we have a gap in the sequence of numbering. We can fix that with the *notes renumber* command.

```
notes renumber race
notes race
```

```
race:
  1. The other category includes people who specified multiple races
  2. This is a third note
```

We can also search notes with the notes search “string” command.

```
notes search .u
```

```
_dta:
  2. The missing values for havechild and childage were coded using -1 and -2 but were
     converted to .n and .u 24 Sep 2025 17:34
```

3.7 Formatting the display of variables

Formatting data will be more common than you expect. It can be a pain when dealing with numbers in the millions or billions and you lack commas. We can format our data with the *format* command.

3.7.1 Format numerics

Let's get our survey data and list the first 5 observations for id and income

Let's look at the format of income.

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use survey5, clear
describe income
list id income in 1/5
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

(Survey of graduate students)

variable name	storage type	display format	value label	variable label
income	float	%9.0g		Income of student

```
+-----+
| id    income |
+-----+
1. | 1    10500.93 |
2. | 2    45234.13 |
3. | 3    1284355   |
4. | 4    124313.5  |
5. | 5    120102.3  |
+-----+
```

The format is *%9.0g*. We always have *%* in front of our format and *g* is a general way of displaying incomes using a width of nine digits and decides for us the best way to display the values. *g* means general here.

%0g means general - find the best way to show the decimals.

Note: *%g* will change the format to exponent if necessary. Also is usually set to 0 with *g*.

%f means fixed - *w* is the width, *d* is the decimals, and *f* means fixed

`%.fc` means fixed with commas - *w* is the width, *d* is the decimals, *f* means fixed, and *c* means comma

`%.0gc` means general with commas - *w* is the width, setting 0 means Stata will decide the decimals, *g* means general, and *c* means comma

The manual is helpful for formatting: <https://www.stata.com/manuals/dformat.pdf>

Example 1: `format v1 %10.0g` - Width of 10 digits and decimals will be decided.

Example 2: `format v2 %4.1f` - Show 3 digits in `v3` and 1 decimal

Example 3: `format v3 %6.1fc` - Show 4 digits plus the comma plus 1 digit

Let's get more control over the income format and use the `%w.df` format. We want a total of 12 digits with 2 decimals places, which means we have 10 digits on the left side of the “.”

```
format income %12.2f
list income in 1/5
```

```

      |      income |
      |-----|
1.    |    10500.93 |
2.    |    45234.13 |
3.    |  1284354.50 |
4.    |   124313.45 |
5.    |   120102.32 |
      +-----+
```

Notice that we now can see observation #3's decimal places.

If we don't care to see the decimal place (even though it is still there).

```
format income %7.0f
list income in 1/5
```

```

      |   income |
      |-----|
1.    |    10501 |
2.    |    45234 |
3.    |  1284354 |
4.    |   124313 |
5.    |   120102 |
      +-----+
```

We want to see one decimal place

```
format income %9.1f
list income in 1/5
```

```

      |      income |
      |-----|
1. |      10500.9 |
2. |      45234.1 |
3. |     1284354.5 |
4. |      124313.5 |
5. |      120102.3 |
      +-----+
```

Now let's add commas, but we need to add two additional digit widths for the commas and we'll add two decimal places.

```
format income %12.2fc
list income in 1/5
```

```

      |      income |
      |-----|
1. |      10,500.93 |
2. |      45,234.13 |
3. |     1,284,354.50 |
4. |      124,313.45 |
5. |      120,102.32 |
      +-----+
```

3.7.2 Format Strings

Let's use the *format* command with strings.

```
describe kidname
```

variable name	storage type	display format	value label	variable label
kidname	str10	%10s		Name of child

The format is *%10s*, which is a (s)tring of 10 characters wide that is right-justified.

```
list kidname
```

```

      |   kidname   |
      |-----|
1.   |             |
2.   |      Sally  |
3.   | Catherine  |
4.   |             |
5.   |   Samuell  |
      |-----|
6.   |             |
7.   |      Robin  |
8.   |             |
      +-----+

```

If we wanted to left-justify the string, we can add a '-' in between % and #s.

```
format kidname %-10s
list kidname
```

```

      |   kidname   |
      |-----|
1.   |             |
2.   | Sally       |
3.   | Catherine  |
4.   |             |
5.   | Samuell    |
      |-----|
6.   |             |
7.   | Robin      |
8.   |             |
      +-----+

```

3.7.3 Format dates

Dates in Stata are a bit of a pain, so learning how to format the dates will be helpful in the future.

```
list bdays kbdays
```

```

      |      bdays      kbdays |
      |-----|
1.   | 1/24/1961          |

```

```

2. | 4/15/1968 4/15/1995 |
3. | 5/23/1971 8/15/2003 |
4. | 6/25/1973          |
5. | 9/22/1981 1/12/1999 |
   |-----|
6. | 10/15/1973          |
7. | 7/1/1977 5/20/1998 |
8. | 8/3/1976          |
   +-----+

```

Our birthdays are in a MM/DD/YYYY format currently. Let's generate a new variable with the `date` function. The `date` function will convert a string that is in a date format into a Stata date, but it still needs to be formatted. The option "MDY" tells Stata that the string is in the Month-Day-Year format and needs to be converted.

```

generate bday = date(bdays, "MDY")
generate kbdays = date(kbdays, "MDY")

```

Let's list the days.

```
list bdays bday kbdays kbdays
```

```

      |      bdays      bday      kbdays      kbdays |
      |-----|
1. | 1/24/1961      389          . |
2. | 4/15/1968     3027  4/15/1995 12888 |
3. | 5/23/1971     4160  8/15/2003 15932 |
4. | 6/25/1973     4924          . |
5. | 9/22/1981     7935  1/12/1999 14256 |
   |-----|
6. | 10/15/1973     5036          . |
7. | 7/1/1977      6391  5/20/1998 14019 |
8. | 8/3/1976      6059          . |
   +-----+

```

The Stata dates are actually stored as the number of days from Jan 1, 1960. This method is convenient for the computer storing and performing date computations, but is difficult for us to read.

Let's use the `%td` format - for example 01Jan2000

```

format bday %td
list bdays bday kbdays kbdays

```

	bdays	bday	kdays	kday
1.	1/24/1961	24jan1961		.
2.	4/15/1968	15apr1968	4/15/1995	12888
3.	5/23/1971	23may1971	8/15/2003	15932
4.	6/25/1973	25jun1973		.
5.	9/22/1981	22sep1981	1/12/1999	14256
6.	10/15/1973	15oct1973		.
7.	7/1/1977	01jul1977	5/20/1998	14019
8.	8/3/1976	03aug1976		.

Let's use the `%tdNN/DD/YY` format...`NN` is used for 01-12 and `nn` is for 1-12, `DD` for the day 01-31, and `YY` is for the last two digits of the year.

```
format bday %tdNN/DD/YY
list bdays bday kdays kday
```

	bdays	bday	kdays	kday
1.	1/24/1961	01/24/61		.
2.	4/15/1968	04/15/68	4/15/1995	12888
3.	5/23/1971	05/23/71	8/15/2003	15932
4.	6/25/1973	06/25/73		.
5.	9/22/1981	09/22/81	1/12/1999	14256
6.	10/15/1973	10/15/73		.
7.	7/1/1977	07/01/77	5/20/1998	14019
8.	8/3/1976	08/03/76		.

`Mon` is Jan-Dec, and `Month` is January-December.

```
format bday %tdMonth/DD/YY
list bdays bday kdays kday
```

	bdays	bday	kdays	kday
1.	1/24/1961	January/24/61		.
2.	4/15/1968	April/15/68	4/15/1995	12888
3.	5/23/1971	May/23/71	8/15/2003	15932
4.	6/25/1973	June/25/73		.
5.	9/22/1981	September/22/81	1/12/1999	14256

6.		10/15/1973	October/15/73	.	
7.		7/1/1977	July/01/77	5/20/1998	14019
8.		8/3/1976	August/03/76	.	
		+-----+			

We can use a standard Month DD, YYYY with the format `%tdMonth_DD,CCYY`. Where *Month* is the full name of the month, *DD* is our days in digits, and *CC* is Century, such as 19- and 20- and *YY* is 2-digit year, such as -88, -97

```
format bday %tdMonth_DD,CCYY
list bdays bday kbdays kbdays
```

		bdays		bday	kbdays	kbdays	

1.		1/24/1961	January 24,1961	.			
2.		4/15/1968	April 15,1968	4/15/1995	12888		
3.		5/23/1971	May 23,1971	8/15/2003	15932		
4.		6/25/1973	June 25,1973	.			
5.		9/22/1981	September 22,1981	1/12/1999	14256		

6.		10/15/1973	October 15,1973	.			
7.		7/1/1977	July 01,1977	5/20/1998	14019		
8.		8/3/1976	August 03,1976	.			
		+-----+					

Let's use a standard format, but don't use YYYY - it just repeats the 2-digit year twice.

```
format bday %tdNN/DD/YYYY
list bdays bday kbdays kbdays
```

		bdays		bday	kbdays	kbdays	

1.		1/24/1961	01/24/6161	.			
2.		4/15/1968	04/15/6868	4/15/1995	12888		
3.		5/23/1971	05/23/7171	8/15/2003	15932		
4.		6/25/1973	06/25/7373	.			
5.		9/22/1981	09/22/8181	1/12/1999	14256		

6.		10/15/1973	10/15/7373	.			
7.		7/1/1977	07/01/7777	5/20/1998	14019		
8.		8/3/1976	08/03/7676	.			
		+-----+					

Use `%tdNN/DD/CCYY` instead for the desired result.

```
format bday %tdNN/DD/CCYY
list bdays bday kbdays kbdays

label variable bday "Date of birth of student"
label variable kbdays "Date of birth of child"
```

	bdays	bday	kbdays	kbdays
1.	1/24/1961	01/24/1961		.
2.	4/15/1968	04/15/1968	4/15/1995	12888
3.	5/23/1971	05/23/1971	8/15/2003	15932
4.	6/25/1973	06/25/1973		.
5.	9/22/1981	09/22/1981	1/12/1999	14256
6.	10/15/1973	10/15/1973		.
7.	7/1/1977	07/01/1977	5/20/1998	14019
8.	8/3/1976	08/03/1976		.

bday and bdays are redundant and we'll only keep one.

```
drop bday kbdays
save survey6, replace
```

3.8 Changing the order of variables in a dataset

I personally find reordering the order of variables with the `order` command to be useful. This is especially true when working with panel data. I like to order the panel data to have the cross-sectional unit first, such as personal id, firm id, etc. first and then have the time period second, so we have our N and T next to one another.

Let's pull our survey of graduate students and describe our dataset

```
cd "/Users/Sam/Desktop/Econ 645/Data/Mitchell"
use survey6, clear
describe
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell
```

```
(Survey of graduate students)
```

Contains data from survey6.dta

```
obs:      8      Survey of graduate students
vars:     9      11 Mar 2024 14:40
size:    416     (_dta has notes)
```

variable name	storage type	display format	value label	variable label
id	float	%9.0g		Unique identification variable
gender	float	%9.0g	mf	Gender of student
race	float	%19.0g	racelab	* Race of student
havechild	float	%18.0g	havelab	* Given birth to a child?
ksex	float	%15.0g	mfkid	* Sex of child
bdays	str10	%10s		Birthday of student
income	float	%12.2fc		Income of student
kidname	str10	%-10s		Name of child
kday	double	%td		

* indicated variables have notes

Sorted by:

We might want to group our variables with similar types of variables. This can be helpful when you have a large dataset with hundreds of variables, such as the CPS.

```
order id gender race bday income havechild
describe
```

Contains data from survey6.dta

```
obs:      8      Survey of graduate students
vars:     9      11 Mar 2024 14:40
size:    416     (_dta has notes)
```

variable name	storage type	display format	value label	variable label
id	float	%9.0g		Unique identification variable
gender	float	%9.0g	mf	Gender of student
race	float	%19.0g	racelab	* Race of student
bdays	str10	%10s		Birthday of student
income	float	%12.2fc		Income of student
havechild	float	%18.0g	havelab	* Given birth to a child?
ksex	float	%15.0g	mfkid	* Sex of child

```
kidname      str10    %-10s                Name of child
kdbday       double   %td
* indicated variables have notes
```

Sorted by:

The variables that we leave off will remain in the same order as before after the new variables are moved to the left.

With the *before* option, we can move variable(s) before a defined variable. Let's move kidname before ksex

```
order kidname, before(ksex)
describe
```

Contains data from survey6.dta

```
obs:      8                Survey of graduate students
vars:     9                11 Mar 2024 14:40
size:     416              (_dta has notes)
```

variable name	storage type	display format	value label	variable label
id	float	%9.0g		Unique identification variable
gender	float	%9.0g	mf	Gender of student
race	float	%19.0g	racelab	* Race of student
bdays	str10	%10s		Birthday of student
income	float	%12.2fc		Income of student
havechild	float	%18.0g	havelab	* Given birth to a child?
kidname	str10	%-10s		Name of child
ksex	float	%15.0g	mfkid	* Sex of child
kdbday	double	%td		

* indicated variables have notes

Sorted by:

We can move newly created variables with the *before* and *after* options with the *generate* command

```
generate STUDENTVARS = ., before(gender)
generate KIDSVARS = ., after(havechild)
describe
```

(8 missing values generated)

(8 missing values generated)

Contains data from survey6.dta

```
obs:      8      Survey of graduate students
vars:     11      11 Mar 2024 14:40
size:     544     (_dta has notes)
```

variable name	storage type	display format	value label	variable label
id	float	%9.0g		Unique identification variable
STUDENTVARS	double	%10.0g		
gender	float	%9.0g	mf	Gender of student
race	float	%19.0g	racelab	* Race of student
bdays	str10	%10s		Birthday of student
income	float	%12.2fc		Income of student
havechild	float	%18.0g	havelab	* Given birth to a child?
KIDSVARS	double	%10.0g		
kidname	str10	%-10s		Name of child
ksex	float	%15.0g	mfkid	* Sex of child
kday	double	%td		
* indicated variables have notes				

Sorted by:

Note: Dataset has changed since last saved.

3.9 Practice

Let's bring in the CPS use “/Users/Sam/Desktop/Econ 645/Data/CPS/jun23pub.dta”,
clear

1. Generate a new variable from pemlr called employed where employed = 1 if the individual is employed (present or absent) and employed = 0 if the individual is unemployed. The value should be missing if the individual is not in the labor force.
2. Label the variable “Currently employed”.
3. Label the values for 0 “Not employed” 1 “Employed” . “Not in the Labor Force”.
4. Move the variable after pemlr.
5. Generate a date that appends hrmonth (month of interview), the string “12”, and the hyear4 (year of interview). We use 12 because the week of the 12th is the reference period.
6. Now format the date so it is like 06/12/2023

Mitchell, N.M. (2020). *Data Management Using Stata: A Practical Handbook*. (2nd ed.). Stata Press.

UCLA Advanced Research Computing. (2025). *How can I access information stored after I run a command in Stata*. Retrieved from <https://stats.oarc.ucla.edu/stata/faq/how-can-i-access-information-stored-after-i-run-a-command-in-stata-returned-results/>

Wooldridge (2009). *Introductory Econometrics: A Modern Approach*. (4th ed.). South-Western Cengage Learning.